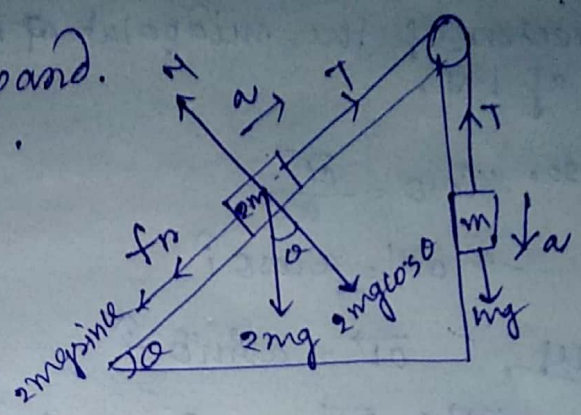
Midsem

(1) Let, 'm' is going downward.
i.e '2m' is going upward.

Let, 'a' be the acc. of the blocks, then



By FBD →

$mg - T = ma$

$\Rightarrow T = m(g \pm a)$ — (1)

$N = 2mg\cos\theta$ — (2)

$T - 2mg\sin\theta - f_r = 2ma$ → frictional force

$\Rightarrow m(g \pm a) - 2mg\sin\theta - \mu_k N = 2ma$ [By (1) & (2)]

$\Rightarrow mg - 2mg\sin\theta - 2\mu_k mg\cos\theta = 3ma$ — (3)

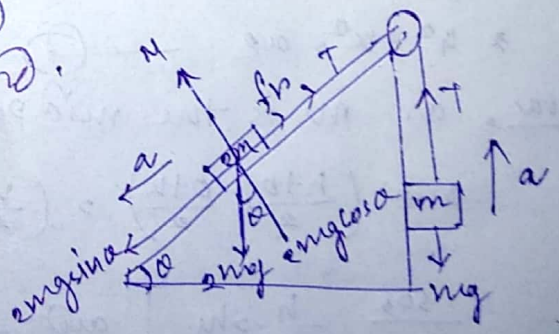
when, $\theta = \theta_0$ → Blocks move with const speed i.e $a = 0$

By (3) →

$mg - 2mg\sin\theta_0 - 2\mu_k mg\cos\theta_0 = 0$

$\Rightarrow \boxed{(1 - 2\sin\theta_0) = 2\mu_k\cos\theta_0}$ → Proved

ans Let, 'm' is going upward and 2m is going downward with acc. a →



$T - mg = ma$

$\Rightarrow T = m(g + a)$

and $2mg\sin\theta - f_r - T = 2ma$

$\Rightarrow 2mg\sin\theta - 2\mu_k mg\cos\theta - m(g + a) = 2ma$

at $\theta = \theta_0$, $a = 0$ $\Rightarrow 2\sin\theta_0 - 2\mu_k\cos\theta_0 = 1$

$\Rightarrow \boxed{(1 - 2\sin\theta_0) = -2\mu_k\cos\theta_0}$ → Also correct.

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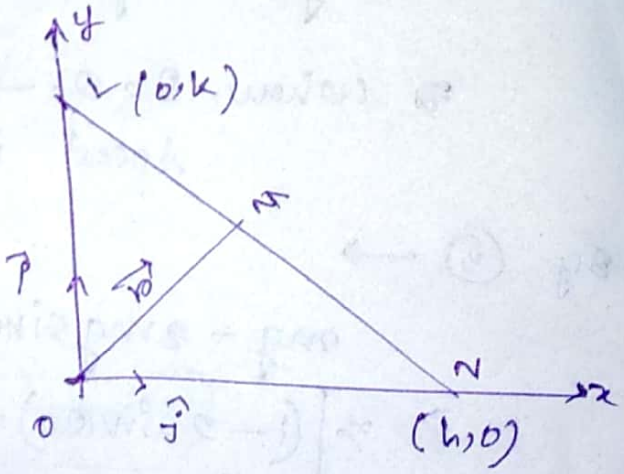
(2)

(a) Let, the co-ordinates,

$N \equiv (h, 0)$ $\quad L \equiv (0, k)$



Now, $|\vec{LN}| = a$

So, $LN^2 = a^2$

$\Rightarrow (h-0)^2 + (0-k)^2 = a^2$

$\Rightarrow h^2 + k^2 = a^2$ —— (1)

Now, as M is the mid point of $\vec{LN}$, so,

$M \equiv \left(\frac{h+0}{2}, \frac{0+k}{2}\right) = \left(\frac{h}{2}, \frac{k}{2}\right) \equiv (h_1, k_1) \rightarrow$ say

So, $\frac{h}{2} = h_1$ | and $\frac{k}{2} = k_1$

$\Rightarrow h = 2h_1$ | $\Rightarrow k = 2k_1$

putting in eqn (1) — $(2h_1)^2 + (2k_1)^2 = a^2$

$\Rightarrow \boxed{h_1^2 + k_1^2 = (a/2)^2}$

So, mid-point M describes the circle of radius $a/2$.

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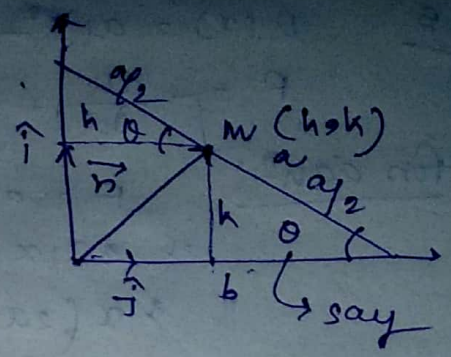
(6) from figure,

we have eqn for the ladder, $x^2+y^2=a^2$



Now $b = a\cos\theta$

$\Rightarrow \frac{db}{dt} = -a\sin\theta\,\dot{\theta}$

$\Rightarrow V_0 = -a\sin\theta\,\omega$ [say, $\dot{\theta} = \frac{d\theta}{dt} = \omega$] — (1)

Now, $M \equiv \left(\frac{a}{2}\cos\theta, \frac{a}{2}\sin\theta\right)$

$\sin\theta = k/\frac{a}{2}$

$\Rightarrow k = \left(\frac{a}{2}\sin\theta\right)$

$\cos\theta = h/\frac{a}{2}$

$\Rightarrow h = \left(\frac{a}{2}\cos\theta\right)$

so $\vec{r} = \frac{a}{2}(\cos\theta\,\hat{j} + \sin\theta\,\hat{i})$

$\Rightarrow \vec{v}_n = \frac{d\vec{r}}{dt} = \frac{a}{2}(-\sin\theta\,\omega\,\hat{j} + \cos\theta\,\omega\,\hat{i})$ — (2)

Now, $\cos\theta = \frac{b}{a}$

so $\sin\theta = (1-\cos^2\theta)^{1/2} = \left(1-\frac{b^2}{a^2}\right)^{1/2}$

Now, (1) $\rightarrow \omega = \frac{V_0}{-a\sin\theta}$

so $\vec{v}_n = \frac{a}{2}\left(\frac{-V_0}{a\sin\theta}\right)(-\sin\theta\,\hat{j} + \cos\theta\,\hat{i})$

$= -\frac{V_0}{2\sin\theta}(-\sin\theta)(\hat{j} - \cot\theta\,\hat{i})$

$= +\frac{V_0}{2}(\hat{j} - \cot\theta\,\hat{i})$

$$\boxed{\vec{v}_n = \frac{V_0}{2}\left(\hat{j} - \frac{b}{\sqrt{a^2-b^2}}\,\hat{i}\right)}$$ Ans

so $|\vec{v}_n| = \frac{V_0}{2}\sqrt{1+\frac{b^2}{a^2-b^2}}$

$$\boxed{|\vec{v}_n| = \frac{V_0 a}{2\sqrt{a^2-b^2}}}$$ Ans

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3. $U(x) = a_1 x^2 - a_2 x^3$

$F = -\frac{dU}{dx} = -(2a_1 x - 3a_2 x^2)$

for eqm^n:

$F = 0$

$\Rightarrow 2a_1 x - 3a_2 x^2 = 0$

$\Rightarrow x(2a_1 - 3a_2 x) = 0$

$\Rightarrow x_0 = 0 \quad \Big| \quad x_0 = \frac{2a_1}{3a_2} = \frac{2 \times 10}{6}$

$= \frac{10}{3}$ — (1)

for stable eqm^n: $\left.\frac{d^2U}{dx^2}\right|_{x_0} > 0$

Now $\frac{d^2U}{dx^2} = 2a_1 - 6a_2 x$

$\left.\frac{d^2U}{dx^2}\right|_{x_0=0} = 2a_1 = 20 > 0$

$\left.\frac{d^2U}{dx^2}\right|_{x_0=\frac{10}{3}} = 2 \cdot 10 - \cancel{6}^{2} \cdot 2 \cdot \frac{10}{\cancel{3}}$

$= 20 - 40 = -20 < 0$

so, $x_0 = 0 \rightarrow$ stable eqm^n pt,

$x_0 = \frac{10}{3} \rightarrow$ unstable eqm^n pt. — (1)

(b) initially, at $x = \frac{a_1}{a_2} = \frac{10}{2} = 5$, $v = 0$

so, $U_i(x=5) = 10(5)^2 - 2(5)^3 = 0$

and $T_i = \frac{1}{2}mv^2 = \frac{1}{2}m(0)^2 = 0$

so, $E_i = U_i + T_i = 0$

at stable eqm^n i.e at $x = 0$, $U_f = 0$

Now $E_f = U_f + T_f = T_f$

By energy conservation, $E_f = E_i$

$\Rightarrow T_f = 0$

$\Rightarrow \boxed{v_f = 0}$

so, velocity at eqm^n pt $= 0$.

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4. Damped angular freq?

(a) $\omega = \sqrt{\omega_0^2 - \frac{b^2}{4m^2}}$

Now $\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{12.5}{0.5}} = \sqrt{25} = 5\ s^{-1}$

Now $\Rightarrow \omega = \omega_0 - \frac{0.2}{100}\omega_0$

$= \omega_0\left(1 - \frac{0.2}{100}\right)$

$= 0.998\omega_0$

so $\omega_0^2 - \frac{b^2}{4m^2} = (0.998\omega_0)^2$

$\Rightarrow \frac{b^2}{4m^2} = \cancel{\omega_0^2[1-0.998\omega_0)^2}\ \omega_0^2 - (0.998\omega_0)^2$

$\Rightarrow b^2 = 4m^2\omega_0^2\ [1 - 0.998^2]$

$= 4m^2\frac{k}{m}\ [1 - 0.998^2]$

$= 4mk[1 - 0.998^2]$

$\Rightarrow \boxed{r = 0.316\ kg\ s^{-1}}$

(b) $A(t) = A_0 e^{-\frac{bt}{2m}}$

$= A_0 e^{-\left(\frac{0.316t}{2\times 0.5}\right)} = A_0 e^{-0.316t}$

(c) Now, $A(t) = A_0 \exp\left(\frac{-bt}{2m}\right)$

$= A_0 \exp\left(\frac{-0.316t}{2\times 0.5}\right)$

$= A_0 e^{-0.316t}$

$\Rightarrow \frac{A(t)}{A_0} = e^{-0.316t}$

$\Rightarrow \frac{1}{2} = e^{-0.316t}$

$\Rightarrow \ln(2) = 0.316t$

$\Rightarrow \boxed{t = \frac{\ln(2)}{0.316} \simeq 2.19\ sec}$

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(d) for critical damping,

$$\omega_0^2 - \frac{b^2}{4m^2} = 0$$

$$\Rightarrow b^2 = 4m\omega_0^2$$

$$\Rightarrow b = 2m\omega_0$$

$$= 2m\sqrt{\frac{k}{m}} = 2\sqrt{mk} = 5\ kg/s^{-1}$$

$$\Rightarrow \boxed{b = 5\ kg/s}$$

if, $b = -ve \rightarrow$ then that will not show any decay or. then $A(t) = e^{-(-b)\frac{t}{2m}}$

$$= e^{+bt/2m}$$

so, $b = +ve$ have.

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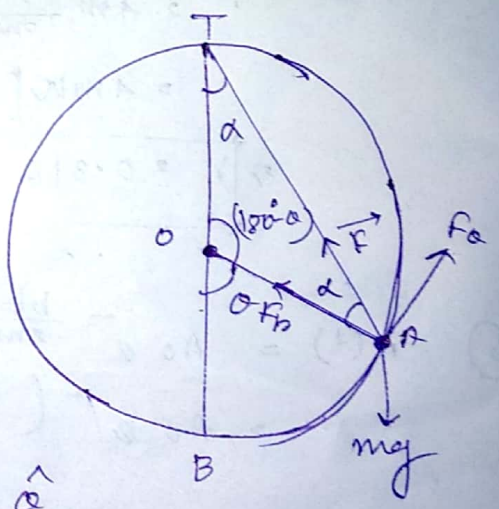
(5) By ~~triangle~~ $\Delta ATO \rightarrow$



(i) $2\alpha + (180° - \theta) = 180°$

$$\Rightarrow \boxed{\alpha = \theta/2}$$

Now, $\vec{F} = F_n\hat{n} + F_\theta\hat{\theta}$

$$= +F\cos\alpha\,\hat{n} + F\sin\alpha\,\hat{\theta}$$

$$= F\cos\theta/2\,\hat{n} + F\sin\theta/2\,\hat{\theta}$$

Now, $d\vec{r} = dr\,\hat{n} + r\,d\theta\,\hat{\theta}$

$= 0 + r\,d\theta\,\hat{\theta}$ [as radius is fixed, so, $dr = 0$]

Now, $dW = \vec{F}\cdot d\vec{r}$

$$\Rightarrow W = \int_{\theta_1}^{\theta_2} \vec{F}\cdot d\vec{r}$$

$$= \int_{\theta_1}^{\theta_2} RF\sin\theta/2\,d\theta$$

$$= 2FR\left[-\cos\theta/2\right]_{\theta_1}^{\theta_2}$$

$$= 2FR\left[\cos\theta_1/2 - \cos\theta_2/2\right] \text{ (Ans)}$$

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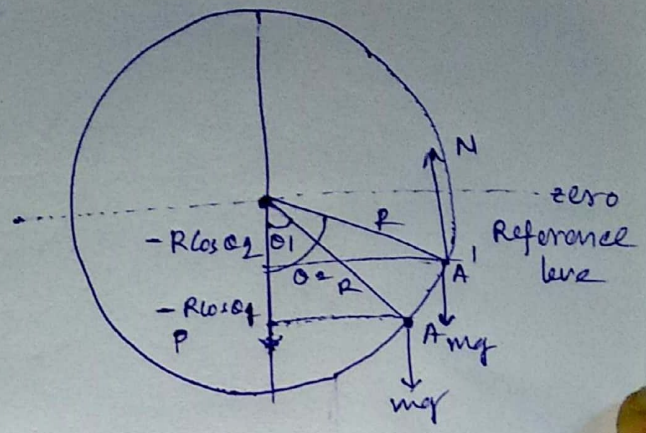
(ii) work done by the weight,

$d\vec{W} = \vec{F}_W \cdot d\vec{r}$



Now $\vec{F}_W = -mg\,\hat{y}$

Now $d\vec{r} = dx\,\hat{x} + dy\,\hat{y}$

Now $dW = -mg\,dy$

$\Rightarrow W_F = -mg\int_{y_1}^{y_2} dy = -mg(y_2 - y_1)$

$= -mg(-R\cos\theta_2 - (-R\cos\theta_1))$

Now $= -mgR(\cos\theta_1 - \cos\theta_2)$ Ans

(iii) By, work-~~energy~~ KE theorem,

$\Delta K = W_F + W_{co} + W_N$ ~~work~~

W_N → work done by normal force

Now, $W_N = 0$ → since N ⊥ to displacement.

So, $\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = 2FR\left(\cos\frac{\theta_1}{2} - \cos\frac{\theta_2}{2}\right) - mgR(\cos\theta_1 - \cos\theta_2)$

$\Rightarrow v = \sqrt{v_0^2 + \frac{1}{m}\left[4FR\left(\cos\frac{\theta_1}{2} - \cos\frac{\theta_2}{2}\right) - 2mgR(\cos\theta_1 - \cos\theta_2)\right]}$

Ans.

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